

VORU, Estonia

WMO: 262490

Lat: 57.8464N

Lon: 27.0194E

Elev: 274

StdP: 14.55

Time Zone: 2.00 (EUE)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.2	(c) -2.1	(d) -13.8	(e) 2.6	(f) -9.3	(g) -6.8	(h) 3.9	(i) -1.2	(j) 17.2	(k) 37.0	(l) 15.4	(m) 35.1	(n) 2.1	(o) 80	(p) 0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.3	(c) 83.2	(d) 66.8	(e) 79.9	(f) 65.0	(g) 76.7	(h) 63.6	(i) 69.0	(j) 78.7	(k) 67.2	(l) 76.2	(m) 65.5	(n) 73.7	(o) 6.6	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	95.7	73.0	63.9	89.9	71.5	62.1	84.1	69.7	33.3	79.0	31.8	76.3	30.4	73.7	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.3	12.9	11.6	-14.5	87.6	9.3	3.0	-21.2	89.8	-26.7	91.6	-31.9	93.3	-38.7	95.5
			-14.5	71.5	8.6	1.9	-20.7	72.8	-25.8	73.9	-30.6	75.0	-36.9	76.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	22.4	23.4	31.0	42.2	53.8	60.0	65.4	62.8	53.7	42.6	35.2	29.5		
	DBStd	16.91	12.59	12.17	8.93	6.94	7.62	6.02	5.68	5.54	5.89	7.52	7.49	9.49		
	HDD50	3844	856	746	589	248	44	4	0	0	30	247	444	636		
	HDD65	7956	1321	1166	1054	684	355	172	64	110	341	694	894	1101		
	CDD50	1514	0	0	0	14	163	303	478	397	141	18	0	0		
	CDD65	150	0	0	0	0	9	22	77	41	1	0	0	0		
	CDH74	1319	0	0	0	1	107	218	643	344	6	0	0	0		
	CDH80	270	0	0	0	0	15	34	150	71	0	0	0	0		
(14) Wind	WSAvg	5.6	6.2	5.9	6.1	5.7	5.3	5.5	4.8	4.6	5.0	5.7	6.0	6.8		
(15) Precipitation	PrecAvg	24.1	1.3	1.1	1.1	1.4	2.0	2.8	3.1	3.2	2.5	2.2	2.0	1.8		
	PrecMax	32.8	3.2	2.5	2.7	4.3	4.5	5.9	7.6	8.4	6.1	4.7	4.1	3.5		
	PrecMin	17.6	0.2	0.2	0.3	0.1	0.4	0.4	0.6	0.4	0.5	0.5	0.6	0.5		
	PrecStd	3.8	0.6	0.6	0.6	1.0	1.0	1.6	1.8	2.0	1.2	1.1	0.8	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	43.0	57.5	71.0	82.1	84.1	87.8	86.0	74.5	62.1	51.7	48.6		
		MCWB	42.8	39.9	44.4	52.3	64.7	66.4	69.1	68.5	61.8	56.0	47.2	46.7		
	2%	DB	41.0	40.4	50.7	65.0	76.8	79.6	84.1	81.4	70.0	57.5	47.9	45.1		
		MCWB	39.2	38.1	41.8	49.7	61.3	62.9	67.6	66.1	59.8	53.7	45.5	42.9		
	5%	DB	38.1	38.4	45.5	60.5	72.5	76.0	80.8	77.8	66.4	55.2	46.3	42.2		
		MCWB	36.5	36.4	39.1	47.3	58.2	61.6	66.5	64.9	58.2	52.0	44.5	40.5		
	10%	DB	36.0	36.5	41.9	55.9	68.4	72.0	77.3	74.1	63.3	53.3	44.5	39.8		
		MCWB	34.7	34.9	37.6	45.0	55.6	59.9	65.0	63.2	56.8	50.5	42.8	38.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.0	40.7	46.3	54.1	67.0	68.8	71.6	71.2	64.5	56.7	48.3	46.8		
		MCDB	44.4	42.3	55.4	66.0	77.9	78.3	82.8	81.5	70.9	59.9	50.1	48.3		
	2%	WB	39.4	38.4	42.3	51.0	63.2	65.7	69.5	68.4	61.5	54.4	46.1	43.2		
		MCDB	40.6	39.9	48.3	61.8	72.8	75.4	79.7	76.8	66.9	57.0	47.5	44.7		
	5%	WB	36.8	36.6	40.1	48.8	60.2	63.3	67.9	66.2	59.4	52.8	44.6	40.6		
		MCDB	37.8	38.0	44.2	58.0	69.1	72.4	77.7	74.2	64.6	55.0	46.1	41.9		
	10%	WB	34.8	35.0	38.1	46.4	57.7	61.3	66.4	64.4	57.6	50.8	42.9	38.5		
		MCDB	35.9	36.3	41.5	54.9	66.2	69.7	75.0	72.5	62.1	52.9	44.3	39.8		
(35) Mean Daily Temperature Range	MDBR		8.3	9.9	13.1	17.6	19.2	16.5	16.3	16.4	13.6	9.8	6.2	6.3		
	5% DB	MCDBR	6.2	7.5	17.7	26.7	25.2	22.2	22.1	22.9	18.9	11.5	7.9	6.6		
		MCWBR	6.0	6.1	11.1	13.8	11.5	9.0	8.5	9.7	10.2	8.2	7.0	6.2		
	5% WB	MCDBR	6.1	6.9	14.9	22.2	21.3	18.2	19.7	18.6	16.0	10.6	7.6	6.1		
		MCWBR	6.1	6.3	10.3	12.7	11.0	9.1	8.5	9.1	9.5	8.3	7.0	6.0		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.307	0.337	0.377	0.371	0.369	0.396	0.393	0.362	0.337	0.316	0.270		
	taud		2.122	2.124	2.170	2.278	2.368	2.405	2.348	2.352	2.418	2.394	2.221	2.120		
	Ebn at Noon		176	225	253	263	274	276	264	256	247	218	160	141		
	Edh at Noon		17	27	34	36	35	34	36	33	27	21	16	12		
(44) All-Sky Solar Radiation	RadAvg		132	351	803	1243	1670	1719	1650	1314	842	375	123	78		
	RadStd		26	63	102	150	178	142	169	132	97	57	18	14		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	-299	-295	N/A	N/A

Nomenclature: See separate page